

Predicting Customer Churn at SyriaTel

Using Machine Learning for Strategic Retention

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Business Understanding

Key Stakeholder

SyriaTel marketing teams

Strategic Objectives

SyriaTel is experiencing customer churn, leading to revenue loss and increased costs associated with acquiring new customers.

The goal is to develop a predictive model that identifies at-risk customers early.



Targeted marketing



Improved customer service



Personalized retention strategies

A key focus is minimizing missed churners, as failing to identify these customers represents lost revenue.

Data Understanding

Data Source

SyrialTel Telecommunications

Historical Data

3,333

Dataset Profile

Records with zero missing values



Features

Behavioural and service-related
variables



Predictive

Indicators ready for churn modeling

Data Preparation

1

Feature Engineering

Removed irrelevant features such as phone number and state to focus on predictive signals.

2

Encoding & Splitting

Encoded categorical variables like international plan and voicemail plan; split data into training and testing sets.

3

Scaling & Integrity

Applied feature scaling for Logistic Regression and prevented data leakage by fitting transformations only on training data.

Modeling Approach

PHASE 1

Baseline

Logistic Regression

- Simple and interpretable
- Provided a performance benchmark
- High accuracy but poor recall for churners

PHASE 2

Improved

Decision Tree

- Captures non-linear relationships
- Significantly improved recall
- Better balance between precision and recall

PHASE 3

Refined

Tuned Decision Tree

- Hyperparameters adjusted to control complexity
- Aimed to reduce overfitting
- No significant recall gain over base model

Logistic Regression

Benchmark Performance

This model established the benchmark and why it was useful for comparison.

It emphasizes that although the baseline achieved high accuracy, it performed poorly at identifying churners.

BASELINE RECALL

0.18

⚠️ Poor Churn Identification

IMPROVED RECALL

0.73

✓ Significant Detection Gain

Decision Tree

Capturing Complexity

This model improved the churn detection objective.

It highlights the Decision Tree's ability to capture non-linear relationships and provide a better balance between precision and recall.

Tuned Decision Tree

Complexity Management

Refinement effort used to manage model complexity and reduce overfitting.

It makes clear that tuning did not meaningfully improve recall beyond the base Decision Tree.



Model Stability

Overfitting Protection Achieved



0.92 Accuracy Maintained

Evaluation

Accuracy



Precision



RECALL



PRIMARY METRIC

F1-score



Why Recall Matters:

Missing a churner means losing a customer without intervention, which directly impacts revenue.

Model Comparison

Performance Matrix

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.86	0.60	0.18	0.27
Decision Tree	0.92	0.73	0.73	0.73
Tuned Decision Tree	0.92	0.73	0.71	0.72

Conclusion

The Decision Tree model provides the best balance between accuracy and recall, making it the most suitable model for SyriaTel's business problem.

Business Recommendations



Risk Identification

Use the Decision Tree model to identify customers at high risk of churn before they leave the network.



Retention Tactics

- Personalized offers
- Proactive customer support
- Service or pricing adjustments



Revenue Protection

Focus on early identification to reduce customer attrition and protect corporate revenue.

Limitations

01

Constraints

Based on historical data; may not capture future behavior changes.

02

External Factors

External factors like competitor pricing are not currently included.

03

Class Imbalance

Slight class imbalance may affect specific model performance segments.

Next Steps



Advanced Modeling

Explore Random Forest and Gradient Boosting to further refine prediction accuracy.



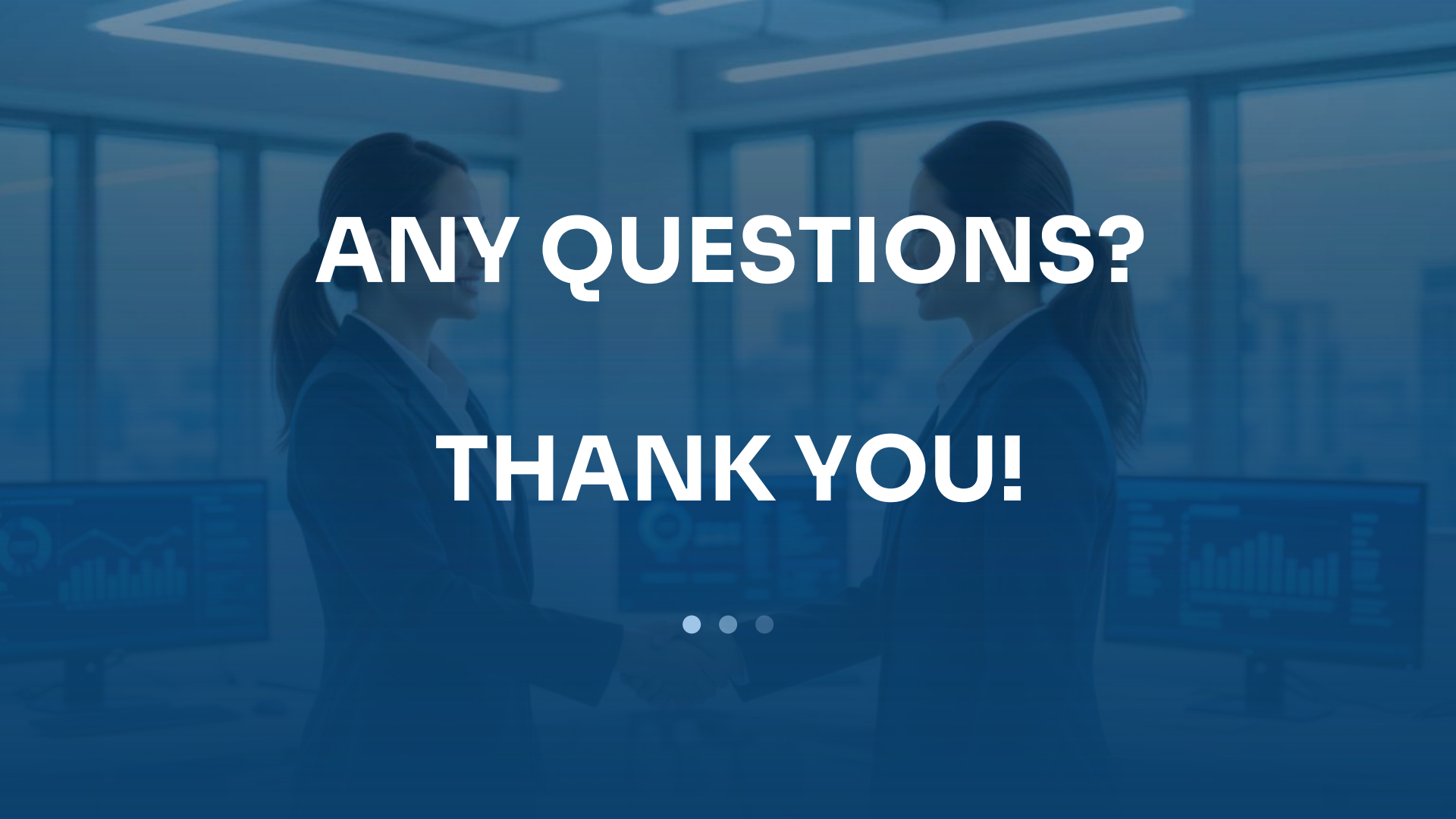
Data Enrichment

Perform feature engineering and incorporate additional data sources for a 360-degree view.



Operationalization

Deploy model for real-time churn monitoring and continuously retrain as behavior evolves.

A blue-tinted photograph of two businesswomen in an office setting, shaking hands. They are both wearing blouses and have their hair in ponytails. In the background, there are large windows and computer monitors displaying charts and graphs. The overall atmosphere is professional and positive.

ANY QUESTIONS?

THANK YOU!

